

```

1  Library IEEE;
2  Use IEEE.std_logic_1164.all;
3
4  Entity ArithUnit is
5      Generic ( N : natural := 64 );
6      Port (
7          A          : in  std_logic_vector(N-1 downto 0);
8          B          : in  std_logic_vector(N-1 downto 0);
9
10         AddnSub     : in  std_logic;
11         ExtWord     : in  std_logic;
12         S           : out std_logic_vector(N-1 downto 0);
13         Cout        : out std_logic;
14         Ovfl        : out std_logic
15
16     );
17 End Entity ArithUnit;
18
19
20
21 Architecture Wrapper of ArithUnit is
22
23     component Adder is
24         Generic ( N : natural := 64 );
25         Port (
26             A, B : in  std_logic_vector(N-1 downto 0);
27             S   : out std_logic_vector(N-1 downto 0);
28             Cin  : in  std_logic;
29             Cout : out std_logic;
30             Ovfl : out std_logic
31         );
32     end component Adder;
33
34     signal A_pre      : std_logic_vector(N-1 downto 0);
35     signal B_pre      : std_logic_vector(N-1 downto 0);
36     signal Cin_pre     : std_logic;
37     signal S_raw       : std_logic_vector(N-1 downto 0);
38     signal Cout_raw    : std_logic;
39     signal Ovfl_raw    : std_logic;
40
41 Begin
42
43
44     gen_b_low : for i in 0 to N/2-1 generate
45         B_pre(i) <= not B(i) when AddnSub = '1' else B(i);
46     end generate;
47
48     gen_b_high : for i in N/2 to N-1 generate
49         B_pre(i) <= '0'      when ExtWord = '1'
50             else not B(i) when AddnSub = '1'
51             else B(i);
52     end generate;
53
54
55     A_pre(N/2-1 downto 0) <= A(N/2-1 downto 0);
56     A_pre(N-1 downto N/2) <= (others => '0') when ExtWord = '1'
57         else A(N-1 downto N/2);
58
59
60
61     Cin_pre <= AddnSub;
62
63
64     AdderInst : component Adder
65         Generic map ( N => N )
66         Port map (
67             A      => A_pre,
68             B      => B_pre,
69             Cin     => Cin_pre,

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70         S      => S_raw,
71         Cout => Cout_raw,
72         Ovfl  => Ovfl_raw
73     );
74
75
76     S(N/2-1 downto 0) <= S_raw(N/2-1 downto 0);
77     S(N-1 downto N/2) <= (others => S_raw(N/2-1)) when ExtWord = '1'
78                          else S_raw(N-1 downto N/2);
79
80     Cout <= Cout_raw;
81     Ovfl <= Ovfl_raw;
82
83 End Architecture Wrapper;

```